Proof-Carrying Diagnostics and Spectral Bounds for Four-Dimensional Wilson $\mathrm{SU}(2)$

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Abstract

The Wilson lattice provides a concrete non-Abelian gauge model, but finite simulation and finite-spacing spectral information must be distinguished from a construction of quantum Yang–Mills theory in four dimensions. We give a unified analysis of a periodic four-dimensional $\mathrm{SU}(2)$ model with reproducible numerical diagnostics and exact rational certificates. An explicit gauge-projected transfer operator connects its positive character expansion with temporal correlators. Using a concentration-uniform Poincaré inequality on the three-sphere, chordal Wasserstein comparison, and the classical Dobrushin method, we reconstruct a volume-uniform bound $\lambda_n/\lambda_0 \le 6\beta$ for the excited gauge-invariant transfer eigenvalues when $0 < \beta < 1/6$. An explicit central-sign Wilson background also yields the opposite, necessary Dobrushin row-sum estimate $\sum_f c_{ef} \ge 9\beta/2$, excluding global single-link row contraction at $\beta \ge 2/9$. We separately prove an exact zero-coupling plaquette covariance and the bias produced by sample-mean centring, give controlled character truncation errors, and exhibit a positive-transfer counterexample to inferring a uniform gap from finite correlators. We give a conditional two-sided physical-block spectral criterion and quantify its sensitivity to transfer-approximation errors. The verification layer replays finite rational consequences, with analytic hypotheses and stochastic diagnostics recorded as distinct claim classes. The strong-coupling estimate relies on established comparison methods and does not establish a continuum mass gap.

1 The question and the principal result

The mass-gap problem asks for a nontrivial quantum Yang–Mills theory on four-dimensional Euclidean space with a strictly positive excitation threshold after its axiomatic construction [4]. A positive spectral estimate on a fixed lattice is a different statement. It can nevertheless be useful when the lattice observable, the transfer operator, and the limiting operation are specified precisely.

We use the Wilson action on a periodic four-dimensional hypercubic lattice with link group $G = \mathrm{SU}(2)$. In our normalization $W(g) = \frac{1}{2}\operatorname{Re}\operatorname{Tr}(g)$, so $W(g) \in [-1, 1]$, and the Gibbs weight is $\exp(\beta \sum_p W(U_p))$. Let $\mathcal{T}_\beta$ denote the actual gauge-invariant, positive Euclidean-time transfer operator defined in Section 2, and let $0 \le \lambda_n \le \lambda_0$ be its eigenvalues. The spatial periods L_1, L_2, L_3 are integers at least two; no condition on their common size is imposed.

Theorem 1.1 (Finite-spacing, gauge-invariant spectral estimate)**.** *For every $0 < \beta < 1/6$, the maximal transfer eigenvalue $\lambda_0 > 0$ is simple, and*

$$0 \le \frac{\lambda_n}{\lambda_0} \le 6\beta < 1 \quad (n \ge 1), \qquad \Delta_{\mathrm{lat}} := \inf_{n \ge 1} -\log\frac{\lambda_n}{\lambda_0} \ge -\log(6\beta) > 0. \tag{1.1}$$

Here $-\log 0 = +\infty$. The bound is independent of the three spatial periods, but is expressed in lattice-time units.

The proof in Section 4 makes explicit a useful route through the unit-quaternion geometry of G. It is a specialization of classical reflection positivity and Gibbs comparison methods [1, 2, 3]. A recent, unrefereed pair of papers discusses the $1/6$ comparison and reports a stronger range $\beta < 2/9$ using a sharp Wasserstein estimate [6, 7]; we neither establish that sharper estimate nor claim priority for the $1/6$ threshold. Section 5 independently constructs an explicit Wilson background which shows why

2/9 is a necessary ceiling for the ordinary unfixed-gauge single-link Dobrushin row-sum method; the ceiling itself is already discussed in the cited unrefereed work. Our contribution is a traceable finite-model reconstruction that places the transfer proof, exact rational enclosures, a rigorous null model, and deliberately limited numerical evidence within one set of verified interfaces.

2 Wilson geometry and the physical transfer operator

Let the Euclidean torus have extents (T, L_1, L_2, L_3) with $T \geq 2$, let $V = L_1L_2L_3$, and associate $U_{x,\mu} \in G$ to each positively oriented link. A reversed link carries $U_{x,\mu}^{-1}$. For the positively oriented plaquette $p = (x; \mu, \nu)$, set

$$U_p = U_{x,\mu}U_{x+\hat{\mu},\nu}U_{x+\hat{\nu},\mu}^{-1}U_{x,\nu}^{-1}, \qquad \mathrm{d}\mu_{\beta,T}(U) = \frac{e^{\beta\sum_p W(U_p)}}{Z_{\beta,T}}\prod_{x,\mu}\mathrm{d}H(U_{x,\mu}), \tag{2.1}$$

where H is normalized Haar measure. Under a site-gauge transformation $U_{x,\mu} \mapsto h_x U_{x,\mu} h_{x+\hat{\mu}}^{-1}$, the plaquette trace and measure are invariant.

At a fixed time, write $U = (U_e)_{e=1}^{3V} \in G^{3V}$ for the spatial positive links and

$$B(U) = \sum_{\text{spatial } p} W(U_p), \qquad |B(U)| \leq 3V, \qquad \mathscr{H} = L^2(G^{3V}, H^{\otimes 3V}). \tag{2.2}$$

Let P be averaging over all site-gauge transformations on that slice. It is an orthogonal projection onto the gauge-invariant subspace $\mathscr{H}_{\mathrm{inv}}$. Define multiplication and temporal convolution on $\mathscr{H}$ by

$$\begin{aligned}(M_\beta f)(U) &= e^{\beta B(U)/2} f(U),\\ (C_\beta f)(U) &= \int_{G^{3V}} \prod_{e=1}^{3V} e^{\beta W(U_e V_e^{-1})} f(V)\,\mathrm{d}H^{\otimes 3V}(V),\\ \mathcal{T}_\beta &= M_\beta P C_\beta P M_\beta \quad \text{on } \mathscr{H}_{\mathrm{inv}}.\end{aligned} \tag{2.3}$$

An overall positive scalar in C_β has no effect on transfer eigenvalue ratios. The gauge-averaged kernel contains one Haar-integrated temporal group variable h_x per spatial site and factors

$$e^{\beta[B(U)+B(V)]/2}\prod_{e=(x,y)} e^{\beta W(U_e h_y V_e^{-1} h_x^{-1})}.$$

Each factor in the product is a timelike Wilson plaquette. Multiplying slabs and integrating their common spatial links reproduces (2.1), including the two spatial half-weights. This is the familiar lattice transfer construction [1], implemented here with independent gauge-transformation and plaquette-factorization checks.

For a bounded real gauge-invariant multiplication observable F, its periodic-time connected correlator has the trace formula

$$C_{\beta,T,F}(t) = \frac{\mathrm{Tr}(\mathcal{T}_\beta^{T-t} F_c \mathcal{T}_\beta^t F_c)}{\mathrm{Tr}(\mathcal{T}_\beta^T)}, \qquad F_c = F - \langle F\rangle_{\beta,T}. \tag{2.4}$$

The positive-transfer structure makes (2.4) nonnegative for the equilibrium finite Gibbs measure. This does *not* force the same sign on every finite sampled and sample-centred Monte Carlo correlator.

3 Characters, exact enclosures and their meaning

Identify G with the unit quaternions S^3 and write $x = W(g) \in [-1, 1]$. The character of the irreducible representation of dimension d is $U_{d-1}(x)$, the Chebyshev polynomial of the second kind. With I_d the modified Bessel function,

$$e^{\beta x} = \sum_{d=1}^{\infty} a_d(\beta) U_{d-1}(x), \qquad a_d(\beta) = \frac{2d}{\beta} I_d(\beta) > 0. \tag{3.1}$$

Schur orthogonality shows that the scalar convolution eigenvalue on dimension-d matrix coefficients is $\gamma_d = 2I_d(\beta)/\beta > 0$. Consequently C_β, its gauge compression $PC_\beta P$, and the positive congruence in (2.3) are self-adjoint and nonnegative. Their rapidly decreasing Peter–Weyl coefficients give compactness and trace class on a finite spatial lattice. The gauge-averaged integral kernel is strictly positive; Perron–Jentzsch therefore gives a simple positive eigenfunction ϕ_0 for λ_0.

The *one-link* fundamental-to-trivial ratio is

$$r_{\mathrm{fund}}(\beta) = \frac{\gamma_2}{\gamma_1} = \frac{I_2(\beta)}{I_1(\beta)}. \tag{3.2}$$

At $\beta = 2$, exact positive-series intervals contain

$$r_{\mathrm{fund}}(2) = 0.433127426722311758\ldots, \qquad -\log r_{\mathrm{fund}}(2) = 0.836723306231588913\ldots. \tag{3.3}$$

Neither number is an eigenvalue ratio of the full interacting gauge-invariant transfer operator. Treating the second as a four-dimensional physical gap would skip the gauge-projected many-link dynamics.

The character expansion also admits a global finite-cutoff certificate. Set $h = \beta/2$, and truncate (3.1) after dimension D. The Bessel series and character norm imply

$$I_d(\beta) = \sum_{k=0}^{\infty} \frac{h^{2k+d}}{k!(k+d)!} \le \frac{h^d}{d!} e^{h^2}, \qquad |U_{d-1}(x)| \le d. \tag{3.4}$$

If $r = h(D+2)/(D+1)^2 < 1$, the omitted pointwise kernel contribution obeys, uniformly for $x \in [-1, 1]$,

$$\left| \sum_{d>D} a_d(\beta) U_{d-1}(x) \right| \le R_D(\beta) := \frac{2e^{h^2}}{\beta} \frac{(D+1)^2 h^{D+1}}{(D+1)!} \frac{1}{1-r}. \tag{3.5}$$

Indeed the displayed bound for each term has successive ratio $h(d+1)/d^2$, which decreases for $d \ge D+1$; the geometric tail begins at the $d = D+1$ term. In the accompanying exact implementation, Bessel sums are enclosed by positive rational intervals and the exponential in (3.5) by an outward rational Taylor estimate. This certifies truncated kernels, not a continuum field theory.

4 A volume-uniform lattice spectral window

The essential improvement over an elementary total-variation comparison comes from measuring a change of a single link in the *chordal* metric $\rho(x, y) = |x - y|$ on $S^3 \subset \mathbb{R}^4$. For $b \in \mathbb{R}^4$, let

$$\mathrm{d}\nu_b(x) = \frac{e^{b\cdot x}}{\int_{S^3} e^{b\cdot z}\, \mathrm{d}\sigma(z)}\, \mathrm{d}\sigma(x), \tag{4.1}$$

where σ is normalized spherical measure. These are the von Mises–Fisher conditional distributions of the Wilson links.

Lemma 4.1 (Uniform spherical Poincaré inequality)**.** *For every $b \in \mathbb{R}^4$ and smooth real $f : S^3 \to \mathbb{R}$,*

$$\mathrm{Var}_{\nu_b}(f) \le \frac{1}{3} \int_{S^3} |\nabla_{S^3} f|^2\, \mathrm{d}\nu_b. \tag{4.2}$$

Proof. By rotation, take $b = \kappa e_1$ with $\kappa \ge 0$, and use polar angle θ from e_1. The angular weight is $w(\theta) = \sin^2\theta\, e^{\kappa\cos\theta}$. Expansion into degree-$\ell$ harmonics on the residual S^2 gives the one-dimensional generator

$$L_\ell = \frac{1}{w}\partial_\theta(w\partial_\theta) - \frac{\ell(\ell+1)}{\sin^2\theta}.$$

The constant mode at $\ell = 0$ has eigenvalue zero. For $\ell \geq 1$ the centrifugal term increases with ℓ. Writing $\psi = \sin\theta\, e^{a\kappa\cos\theta}$, direct differentiation yields

$$-\frac{L_1\psi}{\psi} = 3 + (5a+1)\kappa\cos\theta - a(a+1)\kappa^2\sin^2\theta. \tag{4.3}$$

At $a = -1/5$ this is $3 + \frac{4}{25}\kappa^2\sin^2\theta \geq 3$. The positive supersolution identity

$$\int \left(|u'|^2 + \ell(\ell+1)\csc^2\theta\, u^2\right) w = \int \psi^2 |(u/\psi)'|^2 w + \int (-L_\ell\psi)\psi(u/\psi)^2 w$$

then places every $\ell \geq 1$ eigenvalue above 3. Regular modes vanish as $O(\theta^\ell)$ at a pole; the boundary terms vanish as $O(\theta^{2\ell+1})$.

For a nonconstant zonal eigenfunction g, write $L_0 g = -\lambda g$, $u = g'$, and $d(\theta) = 2\cot\theta - \kappa\sin\theta$. Differentiation yields

$$\mathcal{M}u = -\lambda u, \qquad \mathcal{M} = \frac{1}{w}\partial_\theta(w\partial_\theta) + d'(\theta).$$

In this case

$$-\frac{\mathcal{M}\psi}{\psi} = 3 + (5a+2)\kappa\cos\theta - a(a+1)\kappa^2\sin^2\theta. \tag{4.4}$$

With $a = -2/5$, it is at least $3 + \frac{6}{25}\kappa^2\sin^2\theta$. The smooth zonal derivative vanishes as $O(\theta)$ at each pole, and its ground-state boundary term is $O(\theta^3)$. Thus its eigenvalue is at least 3. All nonconstant spherical sectors have spectral value at least 3, establishing (4.2). □

Lemma 4.2 (Chordal conditional influence)**.** *For all $b, b' \in \mathbb{R}^4$,*

$$W_{1,\rho}(\nu_b, \nu_{b'}) \leq \frac{1}{3}|b - b'|. \tag{4.5}$$

Proof. If f is one-Lipschitz in the chordal metric, its intrinsic spherical gradient has norm at most one almost everywhere. The same holds for $x \mapsto h \cdot x$ when $|h| = 1$. Cauchy–Schwarz and (4.2) give $|\operatorname{Cov}_{\nu_b}(f, h\cdot x)| \leq 1/3$. Differentiating the expectation of f along the linear parameter path $b_s = b + s(b' - b)$ yields

$$\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}_{\nu_{b_s}} f = \operatorname{Cov}_{\nu_{b_s}}(f, (b' - b)\cdot x).$$

Integrate over $s \in [0,1]$ and take the supremum over chordal one-Lipschitz f by Kantorovich duality. □

Proposition 4.3 (Necessary constant at the Haar point)**.** *Any constant K satisfying $W_{1,\rho}(\nu_b, \nu_{b'}) \leq K|b - b'|$ for all $b, b' \in \mathbb{R}^4$ must obey $K \geq 1/4$.*

Proof. For a unit vector $h \in \mathbb{R}^4$, the function $f(x) = h \cdot x$ is chordal one-Lipschitz. At the Haar point $b = 0$, rotational symmetry gives $\mathbb{E}_{\nu_0} f = 0$ and $\operatorname{Var}_{\nu_0}(f) = 1/4$. Differentiating the tilted expectation at $\varepsilon = 0$ yields

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\mathbb{E}_{\nu_{\varepsilon h}} f\Big|_{\varepsilon=0} = \operatorname{Var}_{\nu_0}(f) = \frac{1}{4}.$$

Kantorovich duality implies $W_{1,\rho}(\nu_0, \nu_{\varepsilon h}) \geq |\mathbb{E}_{\nu_{\varepsilon h}} f - \mathbb{E}_{\nu_0} f| = \varepsilon/4 + o(\varepsilon)$ for $\varepsilon \downarrow 0$. Divide by ε and take the limit. □

Thus the available proven constant $1/3$ and the best possible global constant K_* satisfy $1/4 \leq K_* \leq 1/3$. If the upper endpoint were independently reduced to $1/4$, the same 18-incidence row-sum argument would give $q \leq (9/2)\beta$ and the formal window $\beta < 2/9$. Proposition 4.3 is a *necessary* bound on the coefficient, not a proof of that proposed improvement.

Proof of Theorem 1.1. Condition on every link except one fixed four-dimensional positive link e. The six incident Wilson plaquettes are linear in its unit quaternion x_e, so its conditional law is (4.1) with parameter $b_e = \beta \sum_{p \ni e} s_{p,e}$, where each $s_{p,e}$ is a unit-quaternion staple. Inversion and left or right multiplication are chordal isometries. If another link f changes by $\rho(x_f, x'_f)$, then

$$|b_e - b'_e| \le \beta\, m(e,f)\rho(x_f, x'_f),$$

with $m(e,f)$ the number of common plaquette incidences, counted with multiplicity. Exactly six plaquettes meet e; each contains three other link incidences. Hence $\sum_f m(e,f) = 18$, also on side-length-two tori. Lemma 4.2 gives the metric Dobrushin matrix

$$c_{ef} \le \tfrac{\beta}{3} m(e,f), \qquad \max_e \sum_f c_{ef} \le q := 6\beta < 1. \tag{4.6}$$

Write $C = (c_{ef})$ on the full finite set of positive links. For a bounded Lipschitz observable F, write $\delta_e F$ for the supremum of $|F(U) - F(U')|/\rho(U_e, U'_e)$ over pairs of configurations which agree off e. Let F, G both be such observables. The metric Gibbs comparison inequality [3, Corollary 2.6] implies

$$|\operatorname{Cov}(F,G)| \le \sum_{e,f} \delta_e F\, [(I-C)^{-1}]_{ef}\, \delta_f G. \tag{4.7}$$

One can obtain this covariance form by comparing a Gibbs measure with its tilt e^{sG}, dividing the comparison by $|s|$ and taking $s \to 0$. The coordinate diameter of the chordal sphere is two, giving a local tilt perturbation at most $|s|\delta_f G$. Because C has row sum at most q, paths of length at least r have resolvent row sum at most $q^r/(1-q)$. A path from the support of F on time slice zero to its translate at time t requires at least $r = \min(t, T-t)$ plaquette-neighbor steps. Thus for every fixed spatial volume and Lipschitz gauge-invariant F,

$$|\operatorname{Cov}_{\beta,T}(F_0, F_t)| \le A(F,V) q^{\min(t,T-t)}, \tag{4.8}$$

where $A(F,V)$ is finite and independent of temporal period T and separation t.

The Wilson transfer in Section 2 has a smooth, strictly positive integral kernel on the compact spatial configuration manifold. Its nonzero eigenfunctions are smooth, and the Perron eigenfunction ϕ_0 is bounded away from zero. Gauge-invariant functions descend to an orbit space that may have singular strata; smoothness is understood before this quotient. For any normalized excited real eigenfunction ϕ_n, take the bounded Lipschitz observable $F_n = \phi_n/\phi_0$. At fixed t, the period-T transfer correlator converges as $T \to \infty$ to

$$C_{F_n}(t) = \sum_{k \ge 1} |\langle \phi_k, F_n \phi_0 \rangle|^2 (\lambda_k/\lambda_0)^t = (\lambda_n/\lambda_0)^t.$$

Here $\langle \phi_0, F_n \phi_0 \rangle = 0$. Inequality (4.8), first with $T \to \infty$ and then with $t \to \infty$, implies $(\lambda_n/\lambda_0)^t \le A(F_n,V) q^t$; taking tth roots gives $\lambda_n/\lambda_0 \le q$. The constant $A(F_n,V)$ need not be uniform in V: it disappears in this eigenvalue-ratio limit. The case $\lambda_n = 0$ is immediate. □

For example, $\beta = 1/10$ lies outside the elementary total-variation range. Equations (1.1) and (4.6) then give the exact bound

$$\frac{\lambda_n}{\lambda_0} \le \frac{3}{5}, \qquad \Delta_{\mathrm{lat}} \ge -\log(3/5) = 0.510825623765990683\ldots. \tag{4.9}$$

Writing $z = (1-q)/(1+q)$, the finite rational sum $2\sum_{j=0}^{K-1} z^{2j+1}/(2j+1)$ is a strict lower enclosure for $-\log q$. Eight terms at $q = 3/5$ are retained in an exact, tamper-checked receipt.

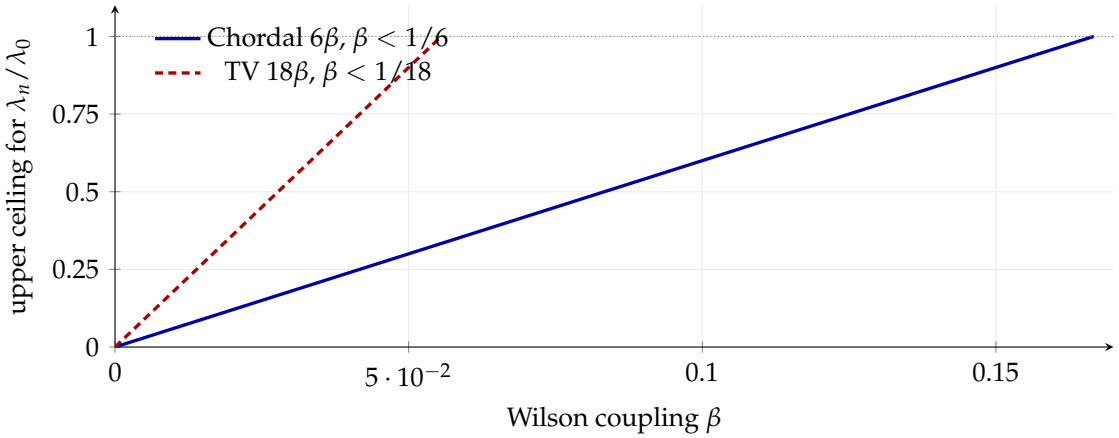


Figure 1: Two proved small-coupling ceilings for the gauge-invariant finite Wilson transfer spectrum. Only the portions below one imply a positive lower gap in lattice units. The curves are analytic bounds, not measured eigenvalues.

5 An intrinsic ceiling for single-link comparison

The previous proof gives a sufficient upper bound on a matrix of conditional influences. We now isolate a necessary lower bound on that same matrix. It prevents the global single-link method from reaching arbitrary Wilson coupling, without asserting any failure of the model's spectral gap.

For distinct positive links e, f, let c_{ef} be the supremum, over configurations differing only at f, of the chordal Wasserstein distance between the two conditional laws of U_e, divided by the chordal distance between those values of U_f. The supremum is taken separately for each f.

Proposition 5.1 (Necessary single-link row-sum ceiling)**.** *On a periodic four-dimensional Wilson* SU(2) *lattice whose four periods are at least three, every positive link e has 18 distinct neighbors f satisfying*

$$c_{ef} \geq \frac{\beta}{4}, \qquad \sum_{f \neq e} c_{ef} \geq \frac{9}{2}\beta. \tag{5.1}$$

Thus the ordinary unweighted single-link Dobrushin criterion $\sup_e \sum_{f \neq e} c_{ef} < 1$ *cannot hold for* $\beta \geq 2/9$.

Proof. Fix $e = (x, 0)$. For each transverse direction $\nu = 1, 2, 3$, the two plaquettes containing e are rooted at x and $x - \hat{\nu}$. Their three other positive links are distinct, and the six plaquettes together involve 18 distinct neighbors because every period is at least three.

Set the three opposite parallel links at $x - \hat{\nu}$ equal to the central element $-I$, and all the other neighboring positive links equal to I. If $q = U_e$, the six incident plaquette contributions are three copies each of $W(q)$ and $W(-q)$, so their sum vanishes identically. The single-link conditional distribution of q is therefore normalized Haar measure on S^3.

Choose any of the 18 neighboring links f and perturb its central value σI to $\sigma \exp(tX)$, with X a unit imaginary quaternion and $\sigma \in \{\pm 1\}$. Only one of the six staples changes. Since the remaining factors are central, the conditional exponential parameter $b_e(t) \in \mathbb{R}^4$ has $b_e(0) = 0$ and $|b_e'(0)| = \beta$. Take $h = b_e'(0)/\beta$ and the chordal one-Lipschitz coordinate function $F(q) = h \cdot q$. At Haar measure $\mathrm{Var}(F) = 1/4$, whence differentiation of the tilted expectation gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \mathbb{E}_{\nu_{b_e(t)}} F \Big|_{t=0} = \frac{\beta}{4}.$$

The chordal distance between the two neighboring link values is $|t| + o(|t|)$. Kantorovich duality proves $c_{ef} \geq \beta/4$. Sum over the 18 distinct neighbors to obtain (5.1). □

The upper estimate $\sum_f c_{ef} \leq 6\beta$ of Section 4 and the lower estimate (5.1) are compatible. The latter does *not* prove that $c_{ef} = \beta/4$, nor the global sharp quarter Wasserstein upper estimate claimed in [7].

It applies to the unfixed-gauge single-link row-sum comparison; block couplings and renormalized methods are not ruled out. The central-sign geometry and its 18 independent link perturbations are tested against the actual Wilson plaquette implementation, with a separate exact arithmetic receipt.

6 An exact Haar null model for temporal correlators

The same physical lattice gives a useful finite-sample control at $\beta = 0$, where all positive-link variables are independent and Haar distributed. For each time slice define the average of its $3V$ spatial plaquettes by

$$X_t = \frac{1}{3V} \sum_{x \in \prod_i \mathbb{Z}/L_i\mathbb{Z}} \sum_{1 \le i < j \le 3} W(U_{(t,x);ij}). \tag{6.1}$$

Proposition 6.1 (Haar covariance and sample-mean bias)**.** *For $L_i \ge 2$ and distinct time slices,*

$$\mathbb{E}_0 X_t = 0, \qquad \mathbb{E}_0[X_t X_u] = \frac{\mathbf{1}_{\{t=u\}}}{12V}. \tag{6.2}$$

If M independent exact Haar-refresh configurations are measured at all T slices, put $N = MT$, $\overline{X} = N^{-1} \sum_{r,t} X_{r,t}$, and

$$\widehat{C}(s) = \frac{1}{N} \sum_{r,t} (X_{r,t} - \overline{X})(X_{r,t+s \pmod T} - \overline{X}).$$

Then, with $v = 1/(12V)$,

$$\mathbb{E}_0 \widehat{C}(0) = v(1 - 1/N), \qquad \mathbb{E}_0 \widehat{C}(s) = -v/N \quad (1 \le s < T). \tag{6.3}$$

Proof. A product of one Haar link and three fixed $\mathrm{SU}(2)$ links is Haar. The real quaternion coordinate $W(g)$ of a Haar group element has mean zero and variance $1/4$. Distinct positively rooted spatial plaquettes have distinct four-link supports, even at spatial period two. To see this, the two direction labels determine the plaquette plane, and the intersections of the starting sites of its positive links determine the root. Some Haar link therefore occurs in exactly one of a distinct plaquette pair; integration over it kills the mixed expectation. Sum the $3V$ diagonal variances, and use disjoint time-slice links, proving (6.2).

For independent zero-mean variables Y_a of variance v and common $\overline{Y}$, direct expansion gives $\mathbb{E}[(Y_a - \overline{Y})(Y_b - \overline{Y})] = v\mathbf{1}_{\{a=b\}} - v/N$. The periodic lag pairs coincide only at $s = 0$. Average this identity to obtain (6.3). □

Adding $1/(12VMT)$ to each lag corrects (6.3) *in expectation* under the independent Haar premise. At $(T, L_1, L_2, L_3, M) = (4, 2, 2, 2, 2)$, the target covariance vector is $(1/96, 0, 0, 0)$, whereas the expected sample-centred vector is $(7/768, -1/768, -1/768, -1/768)$. An exhaustive independent 2^8-case sign regression verifies the algebraic bias formula. The correction must not be applied blindly to positive-β, autocorrelated Monte Carlo data.

7 Finite diagnostics and proof-carrying evidence

The physical simulator uses quaternion $\mathrm{SU}(2)$ links, periodic Wilson plaquettes, gauge-invariant spatial averages, and seeded Metropolis or conditional heatbath updates. A proposed-link action difference touches exactly six plaquettes; tests compare the local update against a direct full-action evaluation on nontrivial fields. Conditional heatbath means are cross-checked against independently evaluated Bessel intervals. These are finite-model implementation checks, not statements about equilibrium in the continuum.

One intentionally short experiment contrasts cold (identity-link) and Haar initializations *using local Metropolis proposals with angle parameter $\varepsilon = 0.3$*, not independent conditional Haar heatbath refreshment. Each trajectory retains all 16 sweeps for seeds 123 and 456. At each sweep the spatial-plaquette average is first computed on each Euclidean time slice and then averaged over *all* time

Coupling; lattice	Cold, early	Cold, late	Haar, early	Haar, late
$\beta = 0,\ 2^4$	0.772843	0.489004	0.039169	0.037911
$\beta = 1/2,\ 2^4$	0.789954	0.564468	0.055750	0.032780
$\beta = 1/2,\ 3^4$	0.797455	0.571755	0.039963	0.081483

Table 1: Recorded short-run spatial-plaquette averages from the retained seeded Monte Carlo traces. They diagnose initialization dependence, not thermalization or a mass gap.

slices; the entries below subsequently average over the two seeds and either sweeps 1–8 or 9–16. The distinct late values show how strongly the cold-start history can persist under this particular small-angle proposal protocol, even when $\beta = 0$.

The verification component called PHYSQED gives each claim a narrowly typed contract. Exact receipts replay the positive Bessel series, rational character tails, the Haar sample-bias formula and rational logarithmic lower sums such as (4.9). The replay checks bounded inputs, fixed claim scope, exact implementation/source digests, and unmodified output fields. Receipts for sampled trajectories retain the complete seeds and measurements. Neither these hashes nor successful tests establish the analytic hypotheses of Lemma 4.1, the infinite-dimensional comparison theorem, or any quantum-field existence axiom; their role is to make each *finite* claim reproducible and resistant to accidental promotion.

8 Why finite decay is not the continuum mass gap

Two logically separate obstacles deserve emphasis. First, a single decaying correlator may place arbitrarily small spectral weight near an arbitrarily small excitation energy. For $n \geq 3$, let

$$T_n = \operatorname{diag}(1, 1/2, 1 - 1/n), \qquad w_n = 1/n,$$

and choose a real self-adjoint observable A_n that couples the vacuum basis vector e_0 to $\sqrt{1 - w_n}\, e_1 + \sqrt{w_n}\, e_2$. Then its normalized vacuum correlator is

$$m_n(t) = \langle e_0, A_n T_n^t A_n e_0 \rangle = (1 - w_n)2^{-t} + w_n(1 - 1/n)^t, \qquad 0 \leq m_n(t) - 2^{-t} \leq 1/n. \tag{8.1}$$

The slow excitation has energy $-\log(1 - 1/n) \to 0$, despite absolute uniform-in-time closeness to 2^{-t}. Both modes couple to A_n. The example is an exact *abstract* positive transfer system, not an SU(2) Wilson counterexample; complete long-time relative asymptotics would reveal its slow mode.

There is also a useful two-sided criterion at a fixed physical time separation. Let $S = \mathcal{T}/\lambda_0$ have simple normalized vacuum Ω, and put $Q = S - |\Omega\rangle\langle\Omega|$ and $r = \|Q\| < 1$. If S is nonnegative and self-adjoint, $\|Q^k\| \leq U < 1$, and a centered observable F obeys $\langle F\Omega, S^k F\Omega \rangle \geq c > 0$ with $\|F\Omega\|^2 \leq A$, then

$$\frac{-\log U}{ak} \leq \frac{-\log r}{a} \leq \frac{-\log(c/A)}{ak}. \tag{8.2}$$

Indeed $F\Omega \perp \Omega$, so positivity and the operator norm give $c \leq Ar^k \leq AU$. The first inequality in (8.2) prevents vanishing physical energy, while the second requires a nonvanishing correlator witness to prevent all excited energies running to infinity. Neither hypothesis has been established along a weak-coupling Wilson continuum sequence. Even uniform physical-energy bounds would not by themselves construct a quantum field theory.

For a normalized excited-sector approximation Q_D with $\|Q\|, \|Q_D\| \leq 1$, the power-telescope identity gives

$$\|Q^k\| \leq \|Q_D^k\| + k\|Q - Q_D\|. \tag{8.3}$$

At ak bounded away from zero, this particular certificate needs a one-step residual of order a or better to retain a fixed contraction margin. It also needs control of vacuum-projector alignment.

Second, a mass in physical units requires a continuum construction and scale setting. If a denotes the lattice spacing, an excitation with lattice energy Δ_{lat} formally has energy Δ_{lat}/a; one cannot

infer a nontrivial continuum spectrum merely by retaining a positive bound at a *fixed* bare β. In four-dimensional asymptotically free SU(2) Yang–Mills theory, the Wilson parameter $\beta = 4/g_0^2$ moves toward the weak-coupling regime as $a \to 0$. The estimate (1.1) applies only at $0 < \beta < 1/6$, so it supplies no spacing-uniform estimate along that trajectory. The finite Peter–Weyl cutoff error also contains the spatial multiplier $\exp(3\beta V)$. To specify this claim, normalize the temporal convolution to have constant-sector eigenvalue one and call it $\widetilde{C}_\beta$. Let Q_D retain representation dimensions at most D on each spatial link and set $\varepsilon_D = \|\widetilde{C}_\beta - Q_D \widetilde{C}_\beta Q_D\|$, bounded above using the positive Bessel series. The projected approximation $\mathcal{T}_{\beta,D} = M_\beta P Q_D \widetilde{C}_\beta Q_D P M_\beta$ then obeys $\|\widetilde{\mathcal{T}}_\beta - \mathcal{T}_{\beta,D}\| \le e^{3\beta V}\varepsilon_D$, where $\widetilde{\mathcal{T}}_\beta = M_\beta P \widetilde{C}_\beta P M_\beta$. The right side is not uniform in volume by that route. A continuum limit would additionally need nontrivial correlation limits and the requisite reflection-positivity and quantum-field reconstruction properties [4, 5].

9 Conclusion and reproducibility

The concrete achievement is a bounded chain from a gauge-invariant Wilson action to its positive transfer operator and a transparent small-coupling spectral inequality, supported by exact rational subcalculations. Independent Haar identities expose a finite-sample centring artefact; short seeded traces demonstrate why equilibration cannot be assumed; and a positive-transfer construction shows why finite correlations are not by themselves spectral certificates. The strongest proved estimate in this account, $\lambda_n/\lambda_0 \le 6\beta$, is restricted to $0 < \beta < 1/6$. The independent central-sign calculation identifies $\beta = 2/9$ as a necessary ceiling for the unfixed-gauge single-link Dobrushin row-sum approach. The conditional physical-block criterion (8.2) exposes two still-open renormalized spectral inputs. This is neither a new proof of Yang–Mills existence nor a positive mass gap in the continuum theory.

The source, finite-lattice implementation, bounded exact receipt generators, committed JSON examples, and regression tests are retained in the `research/yang_mills_su2/` tree of https://github.com/tonynewton79-web/PHYSQED-v1.1. The tests independently differentiate the two Barta identities (4.3)–(4.4) and exercise tamper rejection. Exact link-incidence receipts and a separate physical-block arithmetic module retain the new method-ceiling and conditional scaling distinctions. The next mathematical objective is a block-scale contraction and a nonvanishing correlator witness along the relevant weak-coupling trajectory, followed by continuum reconstruction.

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